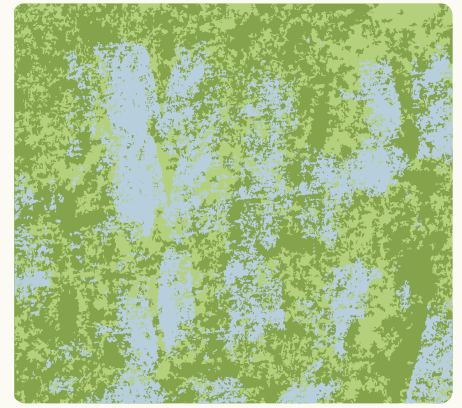
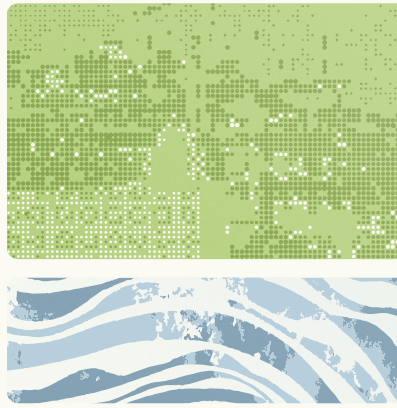


# Meghan Keightley

CS7029 - Introduction



[\*An interactive web version of this document is available to view here\*](#)

## My Background

1. (a) For my bachelors, I studied Software Development at DKIT, which is a 4-year level 8 program, during my studies, I learnt several programming languages, before this I hadn't much experience in programming, I learnt extensive frameworks, such as React and C# for web applications, I also learnt typical DSA in both Java and C++ primarily, I learnt python in reference to data mining and web-scraping. Additionally, I learnt how to build Mobile applications in both Swift & Kotlin for IOS and Android.
1. (b) During my degree, I covered mathematics for computing with topics such as statistics, probability theory, discrete mathematics as well as calculus. I would say I am familiar with mathematic software, such as Mathematica and solving mathematical problems via code.
1. (c) My knowledge towards 3D modelling mainly comes from my semester study abroad, where I learnt how to use blender to stylize & modify 3D models from sketchfab and utilize them in Unity, I feel this is an area I am less familiar with and want to learn more about, so I am excited to cover this area in this course. As for 3D modelling creation overall, I am less knowledgeable in this compared to the other topics covered and will need to focus and apply new information so I can become confident in 3D models. For my [portfolio](#), I utilized them on the web, however these are not my own. I am familiar with [GLTFLoader](#) and aspects of animation via web such as camera lighting, scale etc.
1. (d) As for computer graphics and programming, my course mainly focused on Software engineering so this will be a new avenue for me primarily, but I am quite technical so I should pick it up fairly quick.

## Computer Graphics

2. An area which cites and utilizes computer graphics that I find particularly interesting would be in virtual reality and headsets, I find this interesting due to a technique I learnt known as barrel distortion correction which is essentially a technique designed to trick the human brain which involves consciously distorting and warping the images in a scene, this is done in order to facilitate the graphics for the thick lens that virtual reality headsets have. Together they facilitate the human brain by creating the illusion of immersion which is what makes virtual reality special.
3. I believe Computer graphics in today's world is excellent at providing infinite random worlds in real time for video games and virtual reality experiences which in official terms is known as procedural generation. Instead of building everything by hand, it utilises mathematics and code to generate complex worlds instantly based on what the user does. Traditional art like paintings or sculptures can be limited as an artist can only create so much by hand. Computer graphics has the ability to use computer power to calculate millions of details at once unlike traditional forms of media, which couldn't dynamically calculate and show things such as real-time wind blowing through grass or weather changing it's landscape as you interact with it.

## Longterm Goals

By the end of this course, I would love to create an interactive 3D interpretation of *The Secret Garden* by Frances Hodgson Burnett, which is a favourite novel of mine. I would like to bring the garden to life as an immersive environment, turning the descriptions from the novel into a visual and interactive space. I imagine an overgrown Victorian garden with climbing ivy, roses, trees, a secret doorway and a pond. I would like the garden to feel as though it has been abandoned for years but is gradually coming alive again as the user explores it. I am particularly interested in using 3D modelling, textures and lighting, animation as well as environmental effects such as rain and wind to create this moody but alive atmosphere. Rather than simply recreating the garden from the novel, I would like to create my own interpretation of it and design an experience that captures the feeling of discovery and transformation in *The Secret Garden*. I like the idea of using computer graphics to bring a setting from a book that I love into a space that the user can actually explore.

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